

Task A – Third Normal Form (3NF) Normalization

1. Original Relations (from Phase 2)

Study_Groups(Group_id, Description, Max_participants, Current_participants,
Host_id, Course_id, Location_id, Start_time, End_time, Day, Repeat)

Study_Group_Participant(Participant_id, Group_id)

Users(User_id, Major, User_name, Email)

Enrolled_Courses(Course_id, User_id)

Courses(Course_id, Course_name)

Sections(Section_id, Professor, Semester, Section_of)

Location(Location_id, Building, Room)

2. Functional Dependencies

Study_Groups:

Group_id -> Description, Max_participants, Current_participants,

Host_id, Course_id, Location_id, Start_time, End_time, Day, Repeat

Study_Group_Participant:

(Participant_id, Group_id) -> nothing else

Users:

User_id -> Major, User_name, Email

Enrolled_Courses:

(Course_id, User_id) -> nothing else

Courses:

Course_id -> Course_name

Sections:

(Section_id, Section_of) -> Professor, Semester

Location:

Location_id -> Building, Room

3. 3NF Proof (Relation-by-Relation)

Study_Groups:

Key: Group_id

All attributes depend solely on Group_id.

No partial or transitive dependencies.

In 3NF.

Study_Group_Participant:

Key: (Participant_id, Group_id)

No non-key attributes.

In 3NF.

Users:

Key: User_id

User_id determines Major, User_name, Email.

No non-key -> non-key dependencies.

In 3NF.

Enrolled_Courses:

Key: (Course_id, User_id)

No additional attributes.

In 3NF.

Courses:

Key: Course_id

Course_id -> Course_name.

In 3NF.

Sections:

Key: (Section_id, Section_of)

Both non-key attributes depend on the full composite key.

->In 3NF.

Location:

Key: Location_id

Other attributes depend directly on Location_id.

-> In 3NF.

4. Final 3NF Schema (No Decomposition Required)

Study_Groups(Group_id PK, Host_id FK, Course_id FK, Location_id FK, ...)

Study_Group_Participant(Participant_id PK/FK, Group_id PK/FK)

Users(User_id PK)

Enrolled_Courses(Course_id PK/FK, User_id PK/FK)

Courses(Course_id PK)

Sections(Section_id PK, Section_of PK/FK)

Location(Location_id PK)

All relations are already in 3NF.